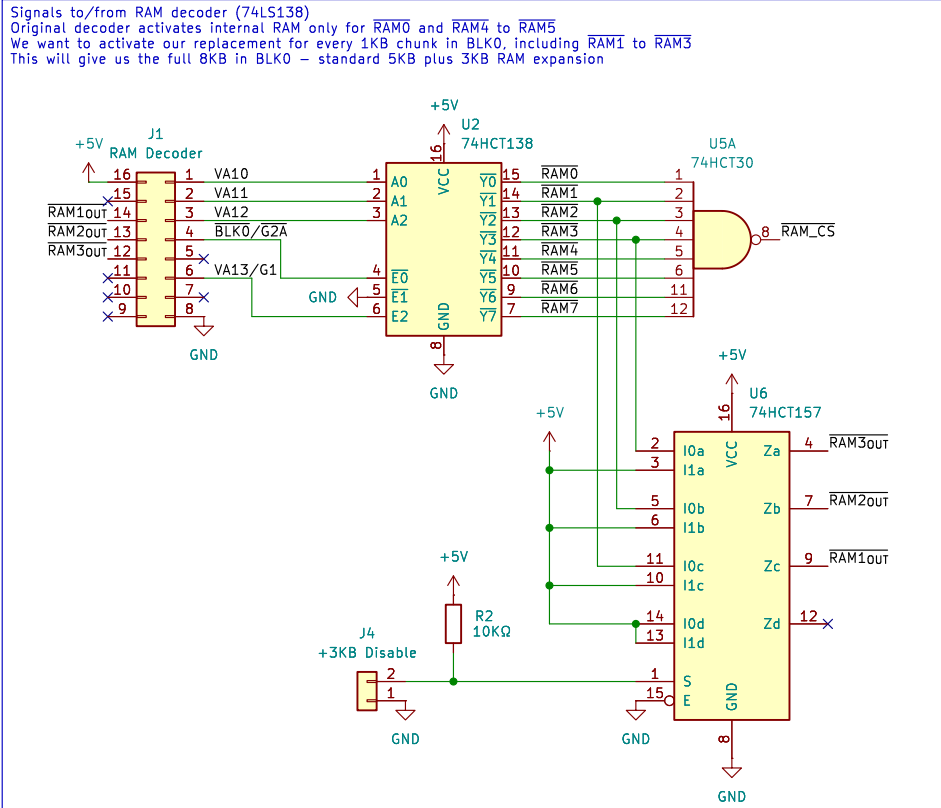
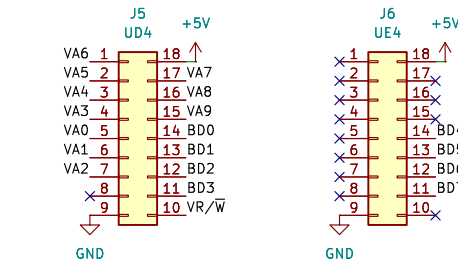




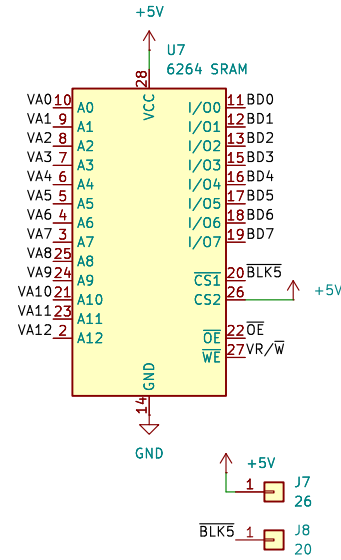
Original 2114 RAM chips are removed from the VIC.
Two are replaced with IC sockets so daughterboard can be installed.



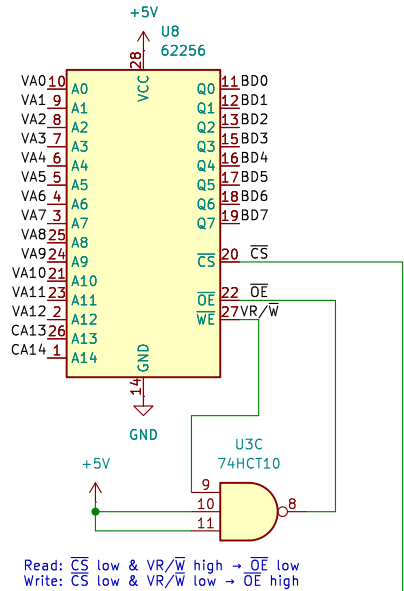
Refer to the "DIP-16 Breakout Board" for the J1 & J2 connection

BLK5 8KB Expansion
Expand to 35KB by piggybacking a 6264 8KB SRAM on top of the 62256 32KB SRAM.

Three pins need special handling:
Pin 1 = lift up, no connection
Pin 20 = lift & connect to BLK5 select
Pin 26 = lift & connect to +5V

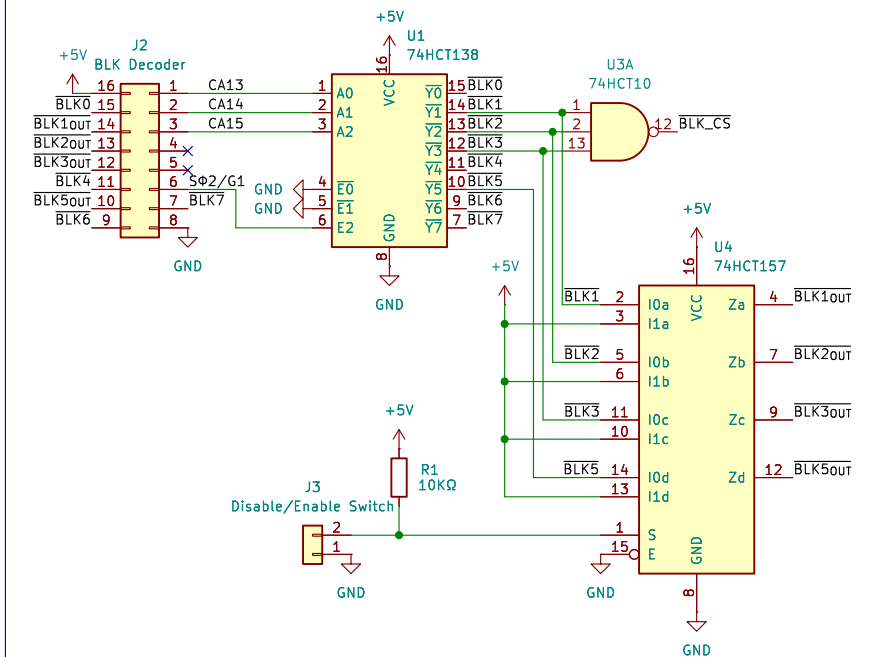


BLK0/1/2/3 32KB RAM Expansion

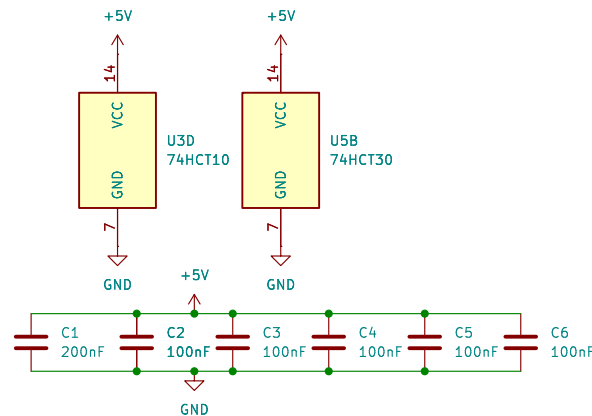


Read: $\overline{CS}$ low & $VR/\overline{W}$ high $\rightarrow$ $\overline{OE}$ low
Write: $\overline{CS}$ low & $VR/\overline{W}$ low $\rightarrow$ $\overline{OE}$ high

Signals to/from BLK decoder (74LS138)
We want to pass BLK0 back so it can be used for BLK0 handling like normal (input to RAM decoder)
We want to capture BLK1 to BLK3 & BLK5 selects and use for our RAM expansion
But we also want to be able to disable the expansion for software compatibility
This can be done via a switch:
* When the switch is high, the BLK1/BLK2/BLK3/BLK5 are passed back to the VIC, disconnecting our RAM expansion from the data bus and passing through to the cartridge port
* When the switch is low, these are blocked which prevents the VIC from disconnecting our RAM from the data bus



Power & Decoupling



U1 & U2	74HCT138 3-to-8 Decoder
U3	74HCT10 Triple 3-Input NAND
U4	74HCT30 8-Input NAND
U5 & U6	74HCT157 Quad 2-Input Multiplexer
U7	62256 32KB SRAM
U8	6264 8KB SRAM
R1	10KΩ Pullup Resistor
C1, C2, C4	100nF Decoupling Capacitor
C3 & C5	200nF Decoupling Capacitor
J1 & J2	2x8 Dupont Female-Female Cables to VIC
J3	Wiring to external SPST switch
J4 & J5	Pair of 1x9 SIL Pin Headers or 2x9 Wirewrap Sockets

www.youtube.com/@Brfff

Brett Hallen

Sheet: /
File: VIC-20_Internal_40KB_RAM_RevB.kicad_sch

Title: Commodore VIC-20 32+3KB RAM Expansion

Size: A3 Date: 27/MAY/2026

KiCad E.D.A. 9.0.9

Rev: B

Id: 1/1